

ID No: _____ Name: _____ Sec No.: _____

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

II SEMESTER 2025-26

CS/ECE/EEE/INSTR F241 Microprocessor Programming & Interfacing

10-02-2026 (Wednesday)

QUIZ-1 (closed book)

A

Max time: 15 Min

Max Marks: 08

Q1. Read the snippet of the program. Data Memory is initialized from offset 012Eh and DS=CS=ES=SS=0863, , DI = 020BH, CX = 0003H, AX = 0000H, BX = 0000H, Assume all flags are cleared initially (i.e '0'). **[1M*8=8M]**

Note: Read the question and then answer- do not spend time understanding the whole program

.MODEL TINY

.DATA

COUNT EQU 2

TITLE DB 'EELAB',0

CHARS DB 41H,42H,43H

FLAGB DB 01H

ARR1 DW 0FFFFH, 7FFFH

ARR2 DW 2 DUP (?)

BIGVAL DD 12345678H

BUFFER DB 4 DUP(?)

.CODE

.STARTUP

MOV SI, OFFSET ARR1

LEA DI, ARR2

MOV CX, COUNT

MOV AX, 0001H

L1:

MOV BX, [SI]

INC WORD PTR [SI]

ADC BX, AX

MOV [DI], BX

ADD SI, 2

ADD DI, 2

DEC CX

JNZ L1

LEA SI, TITLE

MOV AL, [SI]

ADD AL, 20H

MOV BUFFER, AL

.exit

end

Question	Answer (in hex)
(a) What is the value stored at DS:0144 after the program is loaded into memory	12
(b) What is the addressing mode of MOV SI, OFFSET ARR1	Immediate addressing
(c) What is the offset of FLAGB?	0138
(d) What is the value of ARR1 after the execution of the instruction: INC WORD PTR [SI] in the first iteration of L1.	0000
(e) What is the value of SI after the execution of the instruction: INC WORD PTR [SI] in the first iteration of L1.	0139
(f) What is the value of the Carry flag after the execution of instruction ADC BX,AX in the first iteration of the loop L1	1
(g) What is the value of SI (in hex) after the execution of the program	012E
(h) After the execution of the program, what is the word stored at ARR2+2 in hex	8000